

Comparative Evaluation of Convolutional and Transformer Architectures for Lung Cancer CT-Scan Image Classification

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Abstract

Lung cancer remains one of the leading causes of cancer-related mortality worldwide, and the accurate classification of its histological subtypes from computed tomography (CT) imaging is a demanding task even for trained radiologists. This study aims to identify the most suitable deep learning architecture for classifying lung CT-scan images into four classes (Adenocarcinoma, Large Cell Carcinoma, Squamous Cell Carcinoma, and Normal) on a small dataset of approximately 1,000 images, and to deploy the best-performing model as an accessible web application. Four pre-trained architectures were compared using transfer learning: VGG-16, EfficientNetV2, ConvNeXt, and the Vision Transformer (ViT). All models shared an identical classification head and training protocol, and were evaluated on a held-out test set using the weighted F1-score as the primary metric due to class imbalance. VGG-16 achieved the highest weighted F1-score of 0.619, outperforming the more recent and larger architectures, which required more data to generalise effectively. The selected model was subsequently exported and deployed through a FastAPI and Gradio interface hosted on Hugging Face Spaces, with a GitHub Actions workflow providing continuous integration and deployment. The results suggest that, under limited-data conditions, simpler convolutional networks with stronger inductive biases can outperform deeper transformer-based models, while an automated deployment pipeline makes the resulting classifier readily usable by end users.

Keywords: lung cancer; CT-scan; deep learning; transfer learning; web deployment

Introduction

Lung cancer is consistently reported as the leading cause of cancer death globally, accounting for the largest share of cancer mortality in recent international statistics. A major factor behind this high mortality is late diagnosis, as the disease is frequently asymptomatic in its early stages. Computed tomography (CT) is one of the most widely used imaging modalities for detecting pulmonary abnormalities, yet the manual interpretation of CT scans is time-consuming, requires substantial expertise, and is subject to inter-observer variability. Distinguishing between histological subtypes such as adenocarcinoma, squamous cell carcinoma, and large cell carcinoma is particularly challenging because these subtypes can share visually similar radiological characteristics.

Recent advances in deep learning, especially in computer vision, have demonstrated strong potential for assisting medical image analysis. Convolutional neural networks (CNNs) and, more recently, transformer-based architectures have achieved remarkable accuracy across many image classification benchmarks. However, the question of which architecture performs best is highly dependent on the characteristics of the target dataset. Large modern models typically require substantial amounts of labelled data, which is rarely available in specialised medical domains where annotated images are scarce and costly to obtain. In addition, many academic models are never made accessible to end users, remaining as notebooks or weight files that cannot be operated by clinicians or the general public.

This research addresses two objectives. First, it systematically compares four representative architectures, namely VGG-16, EfficientNetV2, ConvNeXt, and the Vision Transformer (ViT), for the four-class classification of lung CT-scan images under a small-data regime. Second, it operationalises the best-performing model by deploying it as an interactive web application, supported by an automated continuous integration and deployment (CI/CD) pipeline. In doing so, the study bridges the gap between model experimentation and practical, end-user-oriented application.

Literature Review/Theoretical Basis

The theoretical foundation of this study rests on transfer learning and the comparison between convolutional and attention-based vision architectures. Transfer learning leverages

representations learned from large-scale datasets such as ImageNet and adapts them to a new, smaller task, which is particularly beneficial when the target dataset is limited in size.

Convolutional Neural Networks and Transfer Learning

VGG-16 is a deep but architecturally simple CNN built from stacked small convolutional filters, and it has long served as a strong transfer-learning baseline. EfficientNetV2 introduces compound scaling and training-aware optimisation to achieve higher accuracy with improved efficiency, while ConvNeXt modernises the standard CNN design by incorporating elements inspired by transformers, aiming to match attention-based models while retaining convolutional inductive biases. These inductive biases, such as locality and translation equivariance, allow CNNs to learn effectively from comparatively fewer examples.

Vision Transformers

The Vision Transformer reformulates an image as a sequence of patches and applies the self-attention mechanism that originated in natural language processing. While ViT can surpass CNNs when trained on very large datasets, its weaker built-in spatial priors mean that it often demands more data or extensive pre-training to generalise well. This trade-off is central to the present study, which evaluates these architectures on a dataset of only around one thousand medical images.

Research Methodology

This study follows an experimental and applied research design, combining a comparative modelling experiment with a software deployment stage. The overall procedure consists of five steps: data preparation, model construction through transfer learning, training, evaluation, and deployment.

The dataset is a publicly available collection of chest CT-scan images grouped into four classes: Adenocarcinoma, Large Cell Carcinoma, Squamous Cell Carcinoma, and Normal, comprising approximately 1,000 images partitioned into training, validation, and test subsets. All input images were converted to RGB, resized to 224 x 224 pixels, and normalised. For VGG-16 and ConvNeXt the pixel values were rescaled by a factor of 1/255, whereas EfficientNetV2 and ViT apply their own internal normalisation; the class indices follow alphabetical order (adenocarcinoma = 0, large cell carcinoma = 1, normal = 2, squamous cell carcinoma = 3).

Each architecture was loaded with weights pre-trained on ImageNet, with the convolutional or transformer backbone frozen and a shared custom classification head attached on top, consisting of a global average pooling layer, batch normalisation, a dense layer of 256 units, dropout of 0.5, a dense layer of 128 units, dropout of 0.3, and a final softmax layer over the four classes. All models were trained with identical callbacks, namely learning-rate reduction on plateau, model checkpointing on the best validation accuracy, and early stopping. The experiments were conducted in Google Colab using a free Tier-4 (T4) GPU with the TensorFlow/Keras framework, and checkpoints were stored on Google Drive. The principal configuration is summarised in Table 1.

Table 1. Summary of the experimental configuration

No.	Component	Setting
1.	Number of classes	4 (Adeno, Large cell, Squamous, Normal)
2.	Dataset size	~1,000 CT-scan images
3.	Input size / colour	224 x 224 pixels, RGB
4.	Transfer learning	ImageNet pre-trained, frozen backbone
5.	Classification head	GAP - BN - Dense(256) - Dropout(0.5) - Dense(128) - Dropout(0.3) - Softmax
6.	Callbacks	ReduceLROnPlateau, ModelCheckpoint, EarlyStopping
7.	Primary metric	Weighted F1-score (+ accuracy, confusion matrix)
8.	Environment	Google Colab T4 GPU, TensorFlow/Keras

Model performance was assessed on the held-out test set. Because the classes are imbalanced, the weighted F1-score was adopted as the primary evaluation metric, complemented by overall accuracy and the confusion matrix. The architecture achieving the highest weighted F1-score was selected for deployment. In the deployment stage, the best model was exported and wrapped in a FastAPI and Gradio interface, then hosted on Hugging Face Spaces; a GitHub Actions workflow automatically synchronises every push to the source repository with the hosted Space, thereby implementing continuous integration and deployment.

Results and Discussion

The four architectures were trained and evaluated under identical conditions. Figure 1 presents the accuracy, precision, recall, and weighted F1-score of each model, and Table 2 reports the corresponding numerical results. VGG-16 obtained the highest weighted F1-score of 0.619 and was therefore selected as the deployment model, while ConvNeXt produced the weakest result and ranked last.

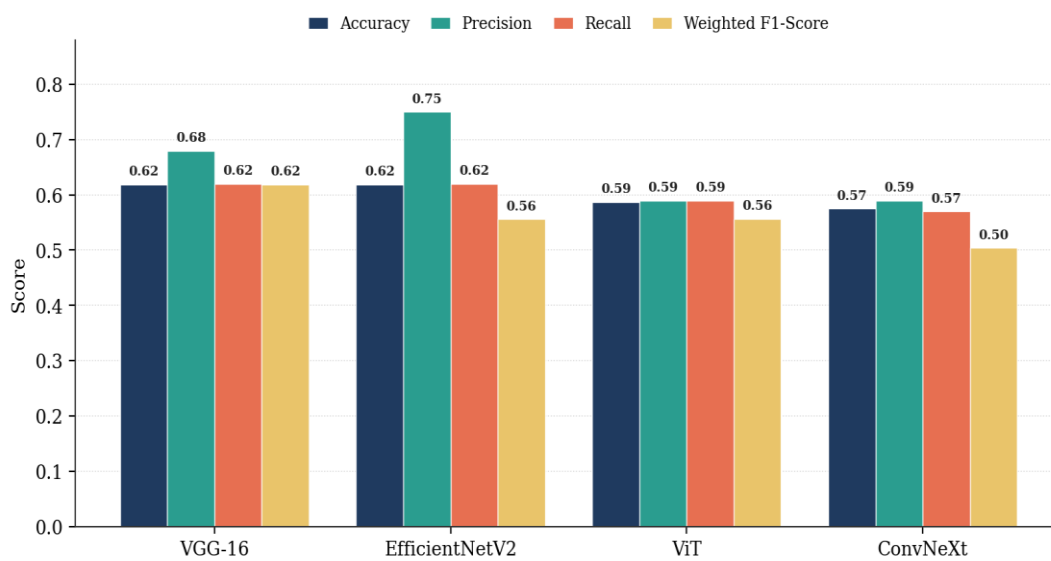


Figure 1. Accuracy, precision, recall, and weighted F1-score of the four architectures.

No	Model Architecture	Accuracy	Precision	Recall	Weighted F1
1.	VGG-16	0.6190	0.68	0.62	0.6193
2.	EfficientNetV2	0.6190	0.75	0.62	0.5555
3.	Vision Transformer (ViT)	0.5873	0.59	0.59	0.5565
4.	ConvNeXt	0.5746	0.59	0.57	0.5039

Table 2. Comparative performance of the four architectures on the test set (weighted precision, recall, and F1).

Two findings stand out. First, although VGG-16 and EfficientNetV2 reach an identical test accuracy of 0.619, VGG-16 attains a clearly higher weighted F1-score (0.619 versus 0.556). This gap is explained by the precision-recall balance of each model: EfficientNetV2 records the highest weighted precision (0.75) yet the lowest F1 among the top group, confirming that it predicts confidently on the majority classes it favours while missing the minority ones. VGG-16, by contrast, balances precision (0.68) and recall (0.62) more evenly across classes, which is preferable for a balanced medical classifier. Second, the two most recent architectures, ConvNeXt and ViT, did not outperform the older VGG-16 on this task.

This outcome is consistent with the theoretical expectation that larger and more data-hungry models require more training data to generalise. On a dataset of only around one thousand images, the stronger inductive biases of a simpler convolutional network such as VGG-16 become an advantage, enabling more data-efficient learning than the weaker spatial priors of the Vision Transformer or the heavily modernised ConvNeXt. An additional contributing factor for ConvNeXt is a preprocessing mismatch: the model expects to perform its own internal normalisation, but it was supplied with inputs that had already been rescaled, resulting in a double-normalisation effect that likely degraded its performance.

Analysis of the confusion behaviour further reveals that the principal difficulty of the task lies not in detecting the presence of cancer but in discriminating between its subtypes. The weakest model, ConvNeXt, separated Normal scans almost perfectly yet frequently confused the cancer subtypes with one another, misclassifying many Large Cell and Squamous Cell cases as Adenocarcinoma. This is clinically plausible, since these tumour subtypes can appear visually similar on CT imaging, and it highlights subtype discrimination as the central challenge for future work.

Following the comparison, VGG-16 was deployed as an end-user web application. Figure 2 illustrates the resulting system architecture. At inference time, an uploaded CT image is preprocessed, passed through the VGG-16 classifier, and converted into a probability distribution over the four classes via a softmax layer. The model is served through a FastAPI and Gradio interface, and an automated GitHub Actions workflow continuously synchronises

the source repository with the live Hugging Face Space, so that every update to the code or model is reflected in the deployed application without manual intervention.

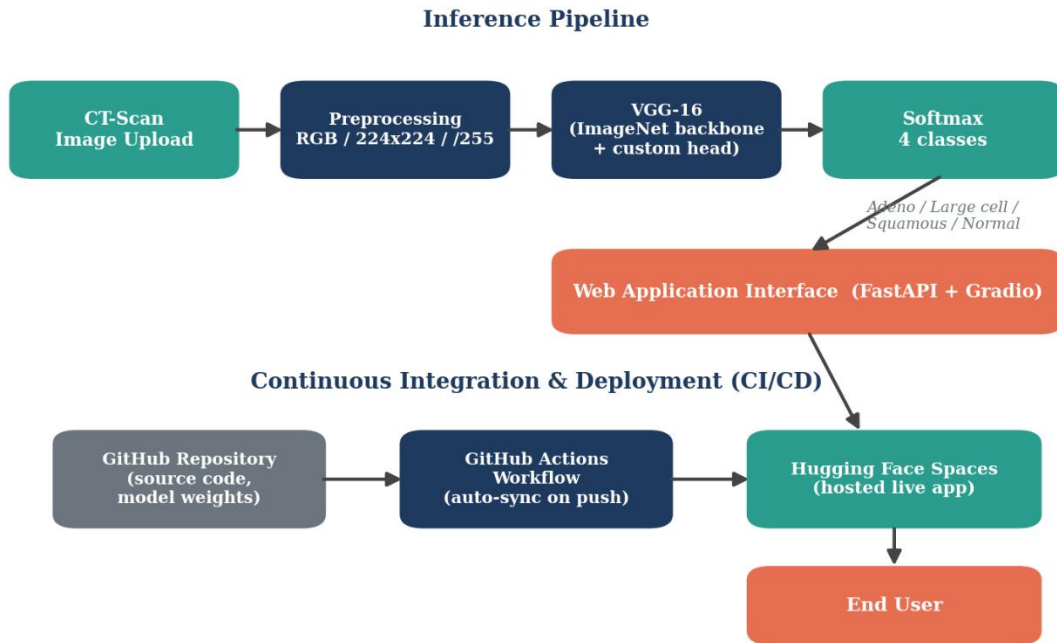


Figure 2. System architecture of the deployment pipeline

Conclusions

This study compared four deep learning architectures (VGG-16, EfficientNetV2, ConvNeXt, and ViT) for the four-class classification of lung CT-scan images and deployed the best-performing model as a web application. VGG-16 achieved the highest weighted F1-score of 0.619 and was selected for deployment, demonstrating that on a small medical dataset a simpler convolutional network with stronger inductive biases can outperform larger and more recent architectures that depend on greater volumes of data. The selected model was successfully delivered as an interactive application hosted on Hugging Face Spaces and supported by an automated CI/CD pipeline, making it directly usable by end users.

Several directions remain for further development. Performance could be improved by enlarging the dataset, applying stronger data augmentation, and selectively fine-tuning the backbone rather than freezing it entirely. Correcting the preprocessing mismatch observed for ConvNeXt would also enable a fairer comparison. Finally, integrating explainability

techniques such as Grad-CAM and pursuing clinical validation would increase the trustworthiness and practical value of the system in real diagnostic settings.

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Authors' contributions

All authors contributed to the conceptualization and design of the study. Aufa Adli Dzil Ikram led the experimentation, developed and trained the models, and assisted with data preparation. Samuel Simatupang built the deployed web application and created the presentation materials. Stefano Viore Bintoro and Mochamat Luthfan Kurniadi were responsible for writing the manuscript. Dhaffa Shaquille Freda Putra revised and refined the presentation. All authors have read and approved the final submitted manuscript.

Availability of data and material

The chest CT-scan dataset used in this study is publicly available on Kaggle. The source code and deployment configuration are available at <https://github.com/m6hhyrz78d-svg/PulmoScan>, and the live demonstration is hosted on Hugging Face Spaces at <https://huggingface.co/spaces/Suizen/Pulmoscan-deploy>.

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